CLINICAL · LARYNX & AIRWAY

Croup and epiglottitis

Acute inflammation of the larynx ranges from the hoarse voice that follows a cold to the child sitting bolt upright, drooling and silent, whose swollen epiglottis may close the airway within the hour. The difference between croup and epiglottitis is one of the oldest lessons in paediatrics and still one of the most important: the barking child can usually be soothed; the drooling child must not be upset, examined or sent to radiology without someone who can secure the airway.

AT A GLANCE

- Croup barks; epiglottitis drools.
- Dexamethasone for croup.
- Don't examine the throat in epiglottitis.
- Secure the airway in theatre.

1 CHAPTER ONE Acute laryngitis

The commonest.

1.1 Causes

- **Infective:** usually viral after a cold (parainfluenza, influenza, rhinovirus), followed by bacterial invasion (pneumococcus, *H. influenzae*, *Moraxella*, streptococci, staphylococci); with measles, chickenpox and whooping cough; candida in inhaled steroid users.
- **Non-infective:** vocal abuse (shouting, cheering), allergy, inhaled irritants, thermal and chemical burns, reflux, and intubation.

1.2 Clinical features

Abrupt hoarseness, sometimes complete loss of voice; pain in the throat after talking; a dry, irritating cough, worse at night; and the features of a cold. Early the epiglottis, aryepiglottic folds, arytenoids and false folds are red and swollen while the true folds stay white, standing out against the redness; later the folds become

red and swollen, with sticky mucus between them and subglottic swelling. Vocal abuse may cause submucosal haemorrhage in a fold. Breathlessness is *not* a feature of simple laryngitis and suggests another diagnosis.

1.3 Treatment

- **Voice rest:** the most important measure; forcing the voice delays recovery and risks haemorrhage and nodules.
- No smoking or alcohol; hydration; steam inhalation.
- Cough suppressants, mucolytics and analgesics.
- Antibiotics only for secondary bacterial infection with fever or purulent sputum.
- Proton pump inhibitors where reflux is suspected.
- Steroids after burns, or for professional voice users who must perform.

Most settle in one to two weeks; hoarseness persisting beyond three weeks needs laryngoscopy.

1.4 Acute membranous laryngitis

Pyogenic membrane on the larynx, often spreading from the pharynx; it must be distinguished from diphtheria by swab and culture.

2 CHAPTER TWO Acute epiglottitis

Supraglottitis.

2.1 Definition and cause

Acute inflammation of the supraglottis (epiglottis, aryepiglottic folds and arytenoids), whose loose mucosa swells until it obstructs the airway; the swollen epiglottis acts like a ball valve. Classically in children of two to seven years from *H. influenzae* type b; since Hib vaccination it is rare in children and now seen mostly in adults, with streptococci, staphylococci and other organisms.

2.2 Clinical features

- **Abrupt onset,** rapid progression over hours, high fever and toxicity (septicaemia).
- **Children:** stridor and difficulty breathing are the presenting features; the child sits upright, leaning forward with the chin out (tripod position), drooling because swallowing hurts, with a muffled voice and usually no cough.
- **Adults:** severe sore throat and painful swallowing out of proportion to a normal-looking oropharynx, a muffled "hot-potato" voice, and tenderness over the hyoid.

DO NOT UPSET THE CHILD

No tongue depressor, no throat examination, no cannula, no lying down, no unaccompanied trip to radiology. Keep the child with the parent, give oxygen without a struggle, and take them to theatre where an experienced anaesthetist and an ENT surgeon ready for tracheostomy can secure the airway under gaseous induction.

2.3 Investigations

- **Lateral soft tissue radiograph** of the neck (only if the patient is stable and accompanied): a swollen epiglottis, the *thumb sign*, with ballooned hypopharynx.
- **Flexible endoscopy** in adults is usually safe and shows a cherry-red, swollen epiglottis and arytenoids.
- Blood cultures (more often positive than surface swabs) and epiglottic swabs after the airway is secure; CT for a suspected epiglottic abscess.

2.4 Treatment

1. **Airway:** intubation (or tracheostomy if intubation fails) in theatre for children and for adults with stridor or rapid progression; selected adults presenting later without stridor can be watched closely in high-dependency care.
2. **Intravenous antibiotics** at once: a third-generation cephalosporin (ceftriaxone or cefotaxime), or ampicillin in sensitive strains.
3. **Steroids** to reduce oedema; humidified oxygen; intravenous fluids.
4. Rifampicin prophylaxis for household contacts of Hib cases, and vaccination.

Intubation is usually needed for 24 to 72 hours, until the epiglottis has shrunk.

3 CHAPTER THREE Croup

Acute laryngotracheobronchitis.

3.1 Cause and pathology

The commonest cause of acute stridor in children, between six months and three years, more in boys and in autumn and winter. Usually *parainfluenza* viruses (types 1 and 2), also RSV, influenza and adenovirus, with secondary bacterial infection. The loose areolar tissue of the subglottis swells inside the rigid cricoid ring; thick secretions and crusts add to the obstruction.

3.2 Clinical features

A day or two of a cold, then hoarseness, a harsh *barking (seal-like) cough* and inspiratory or biphasic stridor, worse at night; fever is usually low or moderate and the child is not toxic. Increasing obstruction shows as suprasternal, intercostal and subcostal recession, tachypnoea and agitation; restlessness, a rising pulse, pallor and cyanosis and finally exhaustion or drowsiness are late, ominous signs.

CROUP AND EPIGLOTTITIS IN CHILDREN

FEATURE	CROUP (LARYNGOTRACHEOBRONCHITIS)	ACUTE EPIGLOTTITIS
Organism	Parainfluenza virus	*H. influenzae* type b
Age	Six months to three years	Two to seven years
Site	Subglottis	Supraglottis
Prodrome	Cold for a day or two	None
Onset	Gradual	Sudden
Fever	Low or moderate	High
General state	Not toxic	Toxic
Cough	Barking	Usually absent
Voice	Hoarse	Muffled

FEATURE	CROUP (LARYNGOTRACHEOBRONCHITIS)	ACUTE EPIGLOTTITIS
Drooling and painful swallowing	Absent	Present
Posture	Any	Upright, leaning forward
Radiograph	Steeple sign on an AP view	Thumb sign on a lateral view
Treatment	Steroids, nebulised adrenaline, humidified oxygen	Airway in theatre, intravenous antibiotics

3.3 Treatment

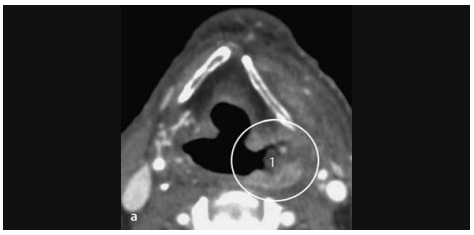
- **Mild:** a single dose of oral dexamethasone (or nebulised budesonide) and observation; most go home.
- **Moderate to severe:** humidified oxygen, steroids and *nebulised adrenaline*, which works within minutes but wears off in about two hours, so observe for rebound; admit.
- **Severe or failing:** intubation in theatre with a tube smaller than usual for age; tracheostomy rarely now.
- Antibiotics only for suspected secondary bacterial infection.

RECURRENT CROUP

A child with repeated attacks, attacks under six months, or croup that will not settle needs microlaryngoscopy and bronchoscopy when well, to look for subglottic stenosis, a subglottic cyst or haemangioma, or a laryngeal cleft.

IMAGING IN THE APP 1 case

Supraglottic swelling on imaging.



Supraglottic and Pharyngeal Tumors
Tumours · 4 images

3.4 Spasmodic croup

Sudden nocturnal attacks of barking cough and stridor in an otherwise well child without fever, settling by morning; often allergic or reflux-related.

4 CHAPTER FOUR Bacterial tracheitis

Croup that is not croup.

A rare, serious bacterial infection (*Staphylococcus aureus*, streptococci, pneumococcus), often following viral croup: high fever, toxicity, stridor and a cough with thick secretions, failing to respond to croup treatment. Bronchoscopy shows purulent tracheitis with oedema, sloughed mucosa and membranes that must be removed, often repeatedly; intubation and intravenous antibiotics; pneumonia frequently follows.

5 CHAPTER FIVE Laryngeal diphtheria

The membrane in the airway.

Usually secondary to faucial diphtheria in unimmunised children under ten. An insidious onset with low fever but marked toxaemia (tachycardia, thready pulse); hoarseness, croupy cough, inspiratory stridor and progressive obstruction; an adherent grey-white membrane on the tonsils, palate and pharynx that bleeds when lifted and extends into the

larynx and trachea; a bull neck. The exotoxin causes myocarditis and palatal, pharyngeal, ocular and diaphragmatic paralysis.

LARYNGEAL DIPHTHERIA

1. Treat on clinical suspicion; take smears and cultures.
2. Antitoxin, 20,000 to 100,000 units intravenously after a test dose, according to severity and duration.
3. Benzylpenicillin or erythromycin for 7 to 10 days.
4. Airway: intubation after laryngoscopy and removal of membrane, or tracheostomy.
5. Strict bed rest for two to four weeks because of myocarditis; cardiac monitoring.
6. Isolation, immunisation of contacts.

CHAPTER SIX

6 Laryngeal oedema

Many causes, one danger.

Oedema collects where the mucosa is loose, in the supraglottis and subglottis, and rarely in the vocal folds.

6.1 Causes

- **Infection:** epiglottitis, croup, tuberculosis, syphilis; nearby quinsy, retropharyngeal abscess and Ludwig's angina.
- **Trauma:** surgery of the tongue and floor of mouth, laryngeal trauma, foreign bodies, endoscopy (especially in children), intubation, burns, caustics and irritant gases.
- **Neoplasm:** ulcerating cancers of the larynx and laryngopharynx.
- **Allergy:** angioneurotic oedema, anaphylaxis; ACE inhibitor and hereditary angioedema.
- **Radiotherapy.**
- **Systemic:** nephritis, heart failure, myxoedema.

6.2 Management

Inspiratory stridor and respiratory distress call for an airway: intubation or tracheostomy. Lesser degrees are treated according to cause: intramuscular adrenaline (0.5 mL of 1:1000 in adults, repeated after five minutes if needed), antihistamines and steroids for allergy; steroids for infective, traumatic and post-radiation oedema; C1 esterase inhibitor or icatibant for hereditary angioedema; stopping the ACE inhibitor.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Croup versus epiglottitis:** the table.
- **How not to examine** a child with suspected epiglottitis, and how to secure the airway.
- **Treatment of croup:** dexamethasone and nebulised adrenaline.
- **Bacterial tracheitis** as croup that fails to respond.
- **Laryngeal diphtheria.**
- **Causes and treatment** of laryngeal oedema.

7.2 Common pitfalls

COMMON PITFALLS

- Using a tongue depressor on a drooling, stridulous child.
- Sending a child with epiglottitis to radiology alone.
- Discharging a child soon after nebulised adrenaline.

- Missing epiglottitis in an adult with severe throat pain and a normal oropharynx.

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IMAGING

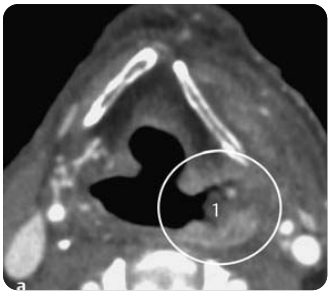
On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · TUMOURS

Supraglottic and Pharyngeal Tumors

- Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and overgrowth of the tumor on the contralateral side.
- Lymphomas (non-Hodgkin) initially might present as unilateral lymphadenopathy and ulcerative lesions.
- Evaluation of systemic disease and lymph nodes in other regions as well as secondary tumors in hypopharynx and larynx.
- Keep in mind that micrometastases may be radiologically occult.



CT with intravenous contrast enhancement, axial. Axial slice at the level of the thyroid cartilage shows a soft-tissue mass (outlined as…